

Plant calendar — Basic design

Document control

Field	Value
Document ID / version	006_BD / 1
Category / system / subsystem	UI / ProductionManagementAI / Master data
Work item / input	WI-010 / approved 006_REQ version 1, DEC-001–007 and RP-01–05
Author / created / updated	Agent / 2026-10-01 / 2026-10-01
Review state	Submitted for review; not implemented

Version	Date	Author	Change
1	2026-10-01	Agent	Initial BD after explicit 006_REQ review approval

System overview and requirements

SCR-006 manages plant-wide weekday rules and dated plant/line exceptions. Its calendar explains working state and effective hours; capacity lookup describes current line/product capability for today/future. It changes no order, due-date, dashboard-window or scheduling behavior.

Requirement	Coverage
REQ-070	Month/scope navigation, state/source display and unavailable dates
REQ-071	Weekly editor, effective dates and retained versions
REQ-072	Plant/line exception editor, manual reasons, remove/restore fallback
REQ-073	Precedence, line-hour fallback and current unit-bearing dated capacity
REQ-074	Role checks, exact validation, concurrency, dirty/unknown states and accessibility
REQ-075	Existing application date and order behavior unchanged

[Approved requirements](#), [decisions](#) and [plan](#) are the sources; acceptance criteria remain authoritative there. RP-01–05 were explicitly accepted in requirements review.

Overall configuration and architecture

Reuse Japanese React/Tailwind UI, central catalog/icons and navigation guard, same-origin ASP.NET Core controllers, Application ports and EF/PostgreSQL. Existing [layering ADR](#) and [cookie/RBAC ADR](#) remain unchanged; [architecture assessment](#) requires no new ADR or technology. Proposed API base `/api/plant-calendar`. Calendar persistence is additive; existing line hours/timing/unit generations remain owned by their master features. Detailed contracts belong to DB/API/FN.

Function list and business flow

Function	Name	Requirements
FN-037	Read monthly effective calendar and selected day	REQ-070, REQ-074
FN-038	Maintain effective weekly patterns with retained revisions	REQ-071, REQ-074
FN-039	Maintain/remove plant and line date exceptions	REQ-072, REQ-074
FN-040	Read current descriptive line/product capacity for a date	REQ-073, REQ-075

1. Admin/Operator opens 「稼働カレンダー」 (Working calendar); select month and plant/line scope and apply. Month defaults to the server's plant month.
2. Inspect a day: working/closed/unavailable, effective hours and winning source.
3. Enter weekly mode to configure weekdays and effective date, or date mode to set an exception for the selected scope/date. Save uses the observed revision.
4. Review confirmation before removing an exception or withdrawing a future weekly version. Successful removal reveals lower-priority configuration.
5. Select current line/product/date for capacity. Display the actual product unit, inputs and current-setting disclaimer; no order allocation or historical result.
6. On invalid/conflicting/uncertain write, retain the draft and reconcile before further mutation. Calendar never automatically retries a write.

Screen list and transitions

All feature modes belong to SCR-006; editor modes are not new screen IDs. Applied month/line/day URL state survives reload; editors initialize from a coherent read. Transient draft edits, confirmation and verification do not create extra routes.

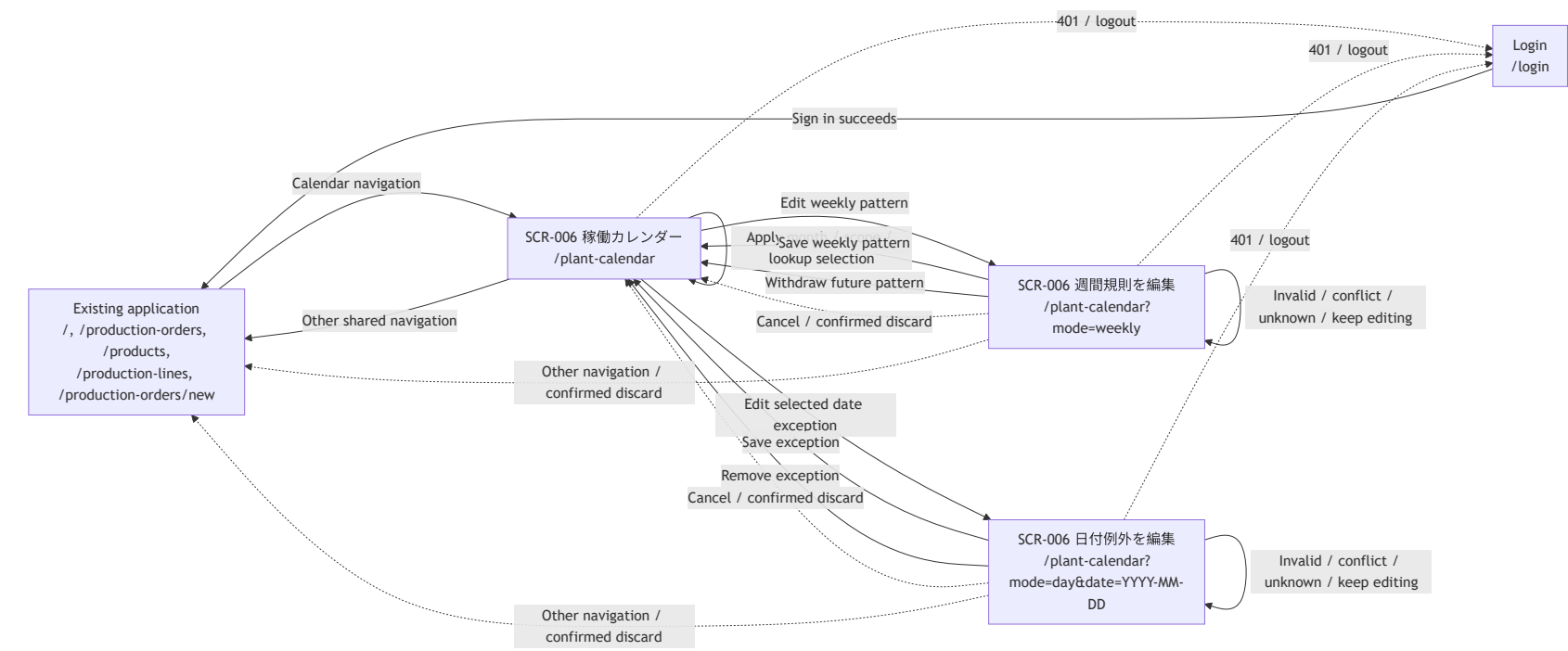
Screen / mode	Route	Entry and exit rule
SCR-006 calendar	<code>/plant-calendar</code>	Shared navigation entry; month/scope stays here; weekly/day actions enter respective modes; other navigation leaves; 401/logout to login
SCR-006 weekly	<code>?mode=weekly</code>	Calendar edit action; save/withdraw/cancel-confirmed-discard returns calendar; invalid/conflict/unknown retains mode; dirty other navigation guarded; 401/logout to login

Screen / mode	Route	Entry and exit rule
SCR-006 date	?mode=day&date=YYYY-MM-DD	Calendar selected-date action; save/remove/cancel-confirmed-discard returns calendar; invalid/conflict/unknown retains mode; dirty other navigation guarded; 401/logout to login
Existing application	/ , /production-orders , /products , /production-lines , /production-orders/new	Existing routes and behavior; Calendar entry opens SCR-006; successful sign-in lands on existing app
Login	/login	401/logout from any mode; successful sign-in returns existing application

The exact transition list below is the diagram's edge inventory.

From	Trigger	To
Existing application routes	Calendar navigation	SCR-006 calendar
SCR-006 calendar	Apply month / scope / lookup selection	SCR-006 calendar
SCR-006 calendar	Edit weekly pattern	SCR-006 weekly mode
SCR-006 calendar	Edit selected date exception	SCR-006 date mode
SCR-006 weekly mode	Save weekly pattern	SCR-006 calendar
SCR-006 weekly mode	Withdraw future pattern	SCR-006 calendar
SCR-006 weekly mode	Cancel / confirmed discard	SCR-006 calendar
SCR-006 weekly mode	Invalid / conflict / unknown / keep editing	SCR-006 weekly mode
SCR-006 date mode	Save exception	SCR-006 calendar
SCR-006 date mode	Remove exception	SCR-006 calendar
SCR-006 date mode	Cancel / confirmed discard	SCR-006 calendar
SCR-006 date mode	Invalid / conflict / unknown / keep editing	SCR-006 date mode
SCR-006 calendar	Other shared navigation	Existing application routes
SCR-006 weekly mode	Other navigation / confirmed discard	Existing application routes
SCR-006 date mode	Other navigation / confirmed discard	Existing application routes
SCR-006 calendar	401 / logout	Login
SCR-006 weekly mode	401 / logout	Login
SCR-006 date mode	401 / logout	Login

From	Trigger	To
Login	Sign in succeeds	Existing application routes



Screen design detail — SCR-006

0-1. Basic information

Item	Design
Route / API family	/plant-calendar / proposed /api/plant-calendar ; exact endpoints deferred
Encoding / channels	UTF-8; one responsive web app; phone emulation is not physical-device proof
Authentication / authorization	Existing cookie; Admin or Operator for every feature read/write

Item	Design
Responsive layout	Existing application small-screen breakpoint; desktop month table, phone agenda; usable at 320 CSS px and 200% zoom
Error fallback	Inline honest error/retry for read; 403 clears feature data and shows forbidden; 401 follows existing auth flow; invalid URL offers explicit reset

0-2. Page metadata

- **0-2-1 Title:** 「稼働カレンダー」 (Working calendar); weekly/day mode may add 「週間規則を編集」 (Edit weekly pattern) / 「日付例外を編集」 (Edit date exception).
- **0-2-2 Base:** no base override.
- **0-2-3 Link:** no screen-specific external stylesheet, font or asset provider.
- **0-2-4 Meta:** inherit existing viewport/document metadata; no public SEO addition.
- **0-2-5 Style:** existing Tailwind classes/tokens; no new inline style block.
- **0-2-6 Script:** existing bundled React route only; no external calendar script.

0-3. URL parameters

Parameter	Meaning	Invalid / change behavior
month	Applied YYYY-MM , default current plant month	Validate before fetch; explicit reset, no silent fabricated month
lineld	Optional selected line identifier; absence means plant scope	Validate identifier/reference; retired line remains readable with restrictions
date	Selected/editor YYYY-MM-DD	Date-only; selected day associated with displayed month; bounded by later API limits
mode	Absent/calendar, weekly or day	Unknown mode rejected with safe return action
productId	Optional capacity selection	Current eligible product for selected line; change invalidates previous result

Changing applied month/scope/lookup invalidates superseded reads and old capacity results. Draft filter values do not silently relabel old applied data. Editor URL is never an authorization source or a promise the date remains editable.

1. Layout and wireframes

The layout samples show 2026-10-12 on L-001: plant maintenance closure overridden by a four-hour line working exception. P-1001 has 0.125 minutes/個; reference capacity is 1,920 個. These are layout samples, not seeded data or runtime evidence.

1

PMAIダッシュボード製造指示一覧製品マスタ生産ライン・工程稼働カレンダーログアウト

2

稼働カレンダー

3

表示月

2026年10月

前の月

次の月

対象ライン

L-001 ― 第1ライン

表示を更新

4

表示：稼働／休み／設定なし ・ 工場日付：2026-10-01（Asia/Tokyo）

5

月	火	水	木	金	土	日
			1 稼働 7.5h	2 稼働 7.5h	3 休み	4 休み
5 稼働 7.5h	6 稼働 7.5h	7 稼働 7.5h	8 稼働 7.5h	9 稼働 7.5h	10 休み	11 休み
12 稼働 4h	13 稼働 7.5h	14 稼働 7.5h	15 稼働 7.5h	16 稼働 7.5h	17 休み	18 休み
19 稼働 7.5h	20 稼働 7.5h	21 稼働 7.5h	22 稼働 7.5h	23 稼働 7.5h	24 休み	25 休み
26 稼働 7.5h	27 稼働 7.5h	28 稼働 7.5h	29 稼働 7.5h	30 稼働 7.5h	31 休み	

6

選択日：2026-10-12

稼働：4時間

適用元：ライン日付例外

ライン理由：臨時稼働

下位：工場例外「保守休業」

過去日は参照のみ

7

週間規則を編集

日付例外を編集

編集可否はサーバーの工場日付で判定

8

日別能力参照：選択ライン L-001

製品：P-1001 ― ブレーキキャリパー

日付：2026-10-12

能力を確認

9

参考能力：1,920 個

適用時間 4時間 × 60 ÷ 0.125分／個

現在の設定による参考値です。予約・実績・過去能力ではありません。

別状態：週間／日付エディター、中央確認、読み込みエラー、競合・結果不明。詳細は本文の各定義。

006_BD version 1 (2026-10-01)

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No.	Region	Notes
1	Shared header/navigation	New 「稼働カレンダー」 entry; central CalendarRange alias proposed, distinct from Factory/Package/ClipboardList; mobile Menu; current-route indication
2	Heading and status	Screen/mode heading; saved/pending/error/conflict/unknown messages with announcement
3	Applied month and scope controls	Previous/next month, month input, plant/line scope and explicit Apply; preserve draft selection independently of applied result
4	State/source legend and plant date	Working, closed and unavailable distinguished by text; server plant-local today/timezone
5	Month days	Desktop semantic month table with date buttons; mobile ordered date list with the same state/source; selected day and configured past read-only information
6	Selected-day details	Effective state/hours, winning source and reason; lower-priority source for explanation; explicit no configuration
7	Editor entry actions	Weekly pattern / selected-date exception; past-date action disabled with explanation; retired-line new-use restrictions
8	Dated capacity selection	Current active line/product and today/future date; explicit request; no historical-capacity input
9	Capacity result / unavailable state	Unit-bearing floor-rounded quantity, hours/minutes basis, current-setting disclaimer; unavailable never fabricated as zero

1

PMAI

メニュー

2 稼働カレンダー

3

表示月

2026年10月

前月

次月

対象ライン

L-001 — 第1ライン

表示更新

4

稼働 / 休み / 設定なし

工場日付：2026-10-01（Asia/Tokyo）

5

2026年10月：日付一覧（抜粋）

10/10（土） 休み：週間規則

10/11（日） 休み：週間規則

10/12（月） 稼働 4h：ライン例外

10/13（火） 稼働 7.5h：週間規則

10/14（水） 稼働 7.5h：週間規則

6

選択日：2026-10-12

稼働：4時間

適用元：ライン日付例外

ライン理由：臨時稼働

下位：工場例外「保守休業」

過去日は参照のみ

7

週間規則を編集

日付例外を編集

編集可否はサーバーの工場日付で判定

8

日別能力参照：選択ライン L-001

製品：P-1001 — ブレーキキャリパー

日付：2026-10-12

能力を確認

9

参考能力：1,920 個

 $4\text{時間} \times 60 \div 0.125\text{分} / \text{個}$

現在設定の参考値。予約・実績ではありません。

過去の日付には能力を表示しません。

別状態：週間／日付エディター、中央確認。

読込・空・エラー・競合・結果不明は本文参照。

配置例です。実装・稼働の証拠ではありません。

The same legend numbers 1–9 apply. The phone layout uses ordered date rows rather than shrinking seven columns; all effective state/source/details remain available. Missing visual states (editors, confirmation, loading, empty, forbidden, conflict and unknown) are specified below and in the SVG note; fuller mockups belong to SPD.

2. Content block definition

None: no CMS, remote holiday content, external identity or public text provider. All Japanese UI strings belong to the existing centralized catalog.

3. Screen item definition

Field / region	Label or meaning	Control / I/O	Initial / source	Conditions and constraints
F-01 / 3	表示月 (Month)	Month input, previous/next, I/O	Applied URL or server plant month	Apply validates month; no write
F-02 / 3	対象ライン (Line scope)	Plant/line selector, I/O	Applied line/plant scope; current server reference	Active line for new use; existing retired reference readable
F-03 / 5–6	Selected day, state, source and reason	Date buttons/list, output details	Coherent month/day read	Past read-only, unavailable explicit; selected date highlighted
F-04 / weekly	有効開始日 (Effective from)	Date input, input	Server plant today for new definition; retained version when inspecting	Today/future for mutation; old effective result cannot change
F-05 / weekly	稼働曜日 (Working weekdays)	Seven labeled checkboxes, input	Effective definition; first Mon–Fri default	Boolean weekday set; all closed allowed, not guessed as invalid
F-06 / weekly	Retained/current/future versions	Definition list, output/select	Server effective versions	Past read-only; future correction/withdrawal preserves earlier records
F-07 / date	対象・日付 (Scope/date)	Read-only labels on existing exception; selection for new, I/O	Captured selected scope/date	Existing identity is not silently moved; new scope/date creates its own exception
F-08 / date	稼働状態 (Working state)	Working/closed choice, input	Existing override or explicit new choice	Closed implies zero effective hours; no contradictory explicit hours
F-09 / date	ライン設定を使用 (Use line hours)	Labeled checkbox, input	True if working exception has no explicit hours	Working only; omission uses current line hours, not lower exception hours
F-10 / date	稼働時間 (Working hours)	Exact decimal text input, input	Existing explicit text or blank	Required only when working and not inheriting; >0, <=24, <=3 decimals
F-11 / date	理由 (Reason)	Text input, input	Existing reason or empty	Optional business note; technical bounds in DB/API; never HTML execution
F-12 / editors	保存／キャンセル (Save/Cancel)	Buttons, input	Save disabled until a valid changed draft	Preserve input on invalid/conflict; pending prevents double write
F-13 / editors	削除／予約規則取消 (Remove/withdraw)	Confirmation action, input	Existing editable exception/future definition	Captured identity/version; no past delete; no mutation on Cancel/Escape

Field / region	Label or meaning	Control / I/O	Initial / source	Conditions and constraints
F-14 / 8	Line/product/date capacity query	Selector/date/request, input	Current selected active line, eligible product and today/future	No read result reused after selection changes; no past capacity
F-15 / 9	参考能力 (Reference capacity)	Output	Effective hours/current confirmed timing/unit snapshot	Unit-bearing floor-rounded quantity; unavailable explicit; no cross-product total
F-16 / 2	Messages and verification	Status/alert and guarded read action, output/input	Result of current action	Conflicts/unknown preserve draft; input values not proof a save succeeded

Technical maximum lengths/date bounds and exact payload transport belong to DB/API/DD. Hours are not floating-point-coerced or padded to imply precision. Internal revision tokens are not user-editable business fields.

4. Item value mapping and effective resolution

Situation	Display / behavior
First weekly activation	Mon–Fri working from plant-local activation date; earlier unavailable when no recorded applicable rule
Line date exception exists	Wins wholly over plant exception and weekly pattern; may reopen/close day
Otherwise plant exception exists	Wins over effective weekday definition
Otherwise effective weekly definition exists	Its weekday state applies
No applicable recorded rule	「設定なし」 (Configuration unavailable), no invented working state/hours/capacity
Winning working rule has no explicit hours	Selected line's current WI-009 hours; plant-only view says line-dependent
Winning rule closes day	Effective hours zero; usable line/product capacity zero, with nonworking reason
Usable working line/product	Hours × 60 ÷ minutes/unit, round down whole discrete units or up to three decimals kg/m
Retired/incompatible/stale-unit configuration	Calendar retained reference may remain; capacity unavailable, not fabricated zero
Past selected date	Calendar definition where recorded; no capacity or edit action

A line working exception with inherited hours uses the line baseline even if a lower-priority plant exception supplied different hours. No partial merging of a lower-priority exception's working state/time into the winning exception.

5. Validation definitions

Rule	Timing	Error outcome
Admin/Operator	Every read/write, independent of route/UI	401 existing auth flow; 403 no feature data
Valid date/month/scope/reference	Before query/save and authoritative server check	Field or page error, no synthetic defaults
Today/future mutation; future-only weekly withdrawal	At authoritative mutation boundary using plant clock	Past-date/future-version violation, preserve draft and refresh editable context
One current exception per scope/date; unambiguous weekly effective rule	Atomic save; DB consistency in later design	Duplicate/conflict identified, no hidden overwrite
Exact hours and working/closed consistency	On submit and server	Field error and first-invalid focus; no implicit zero/default
Stale observed version	Before transition/removal	Conflict, no partial write; explicit reload with dirty confirmation
Active reference/current timing generation for new use or capacity	Current coherent read/mutation validation	Unavailable/rejected, no automatic conversion/reconfirmation
Bounded reason/query/payload	Before processing; bounds in API design	Explicit rejection and generic safe diagnostic

6. Item events

Event	Result
Apply month/scope	Validate, update applied URL, cancel obsolete reads; announce updated month
Select day	Show matching effective state/source; keep the day focus reachable
Enter editor	Load observed definition/version and server date; retain context for return
Toggle closed/use-line-hours	Update draft explicitly; clear contradictory explicit hours, retain dirty indication
Save succeeds	Return calendar with preserved month/scope/day; refresh related state and announce success
Remove exception / withdraw future definition	Centered confirm using captured target/version; submit once; reveal fallback on success
Cancel or other navigation with dirty draft	Confirm discard; Escape/Cancel retains current values; confirm leaves to intended destination
Read retry	Explicit read retry; no write replay; old failed data not labeled current
Unknown write verification	Cancel remains available; block further mutation, read current state and require reconciliation

7. External identity linkage

None. Existing cookie login only; no new SSO/payment/provider mapping.

Success and exception flows

Flow	Behavior / resulting state
Month/day read	Matching snapshot shown; unavailable dates honest; errors explicit
Weekly/exception save	One atomic transition with retained revisions; refresh calendar after success
Removal/withdrawal	Confirmation centered in viewport; target/version captured; fallback revealed after confirmed success
Invalid input	Remain in mode, retain values; errors announced and focused
Conflict	Preserve draft; no automatic merge; guarded reload/reconciliation
Pending	Prevent duplicate mutation and accidental destructive leave; show processing state
Unknown committed outcome	Do not announce success or failure as certain; keep draft, block replay, allow cancel/read verification
Midnight or reference race	Server may reject now-past/newly invalid input; update explanation without reinterpreting history
Dirty navigation	Confirm discard before router/browser history transition; browser exit warning where supported

Data design overview and migration boundary

Calendar owns effective weekly patterns and one current scope/date exception, with prior revisions retained to protect earlier applicable definitions. A line exception references existing line identity. Derived capacity reads existing line hours, product unit/generation and confirmed pair timing, coherently with calendar resolution. No new quantity history or order relation is required. Additive owner-run migration initializes activation explicitly; dates before recorded coverage unavailable. Restricted runtime grants; no runtime DDL, historic backfill, existing line-hours rewrite or live execution in this phase. DB specifies physical constraints, indexing and recovery/forward-fix limits later.

External interfaces and non-functional requirements

No external calendar/provider/job. Bounded same-origin month/scope/version reads, calendar mutations and single line/product/date capacity read; exact endpoint contract, numeric/date serialization and transaction budgets belong to API/FN. Queries must be bounded, cancellation-aware and avoid per-day database roundtrips. Use coherent snapshots and version checks, not fabricated mixed-generation data.

WCAG 2.2 AA design targets: associated labels, text plus color, adequate touch controls, semantic table/native day buttons or complete keyboard pattern, mobile agenda, visible focus, errors/status announcements and 320px/200% reflow without page-wide horizontal overflow. Native fixed confirmation is centered relative to

viewport, traps forward/reverse Tab, starts on safe Cancel, restores trigger focus and supports Escape when safe. Retired/past restrictions have textual reasons. These are design targets, not runtime accessibility certification.

Use bounded operation/outcome spans, count/duration metrics and generic traceId errors; record resolution/validation/transaction outcome without raw date/line/product/reason/credentials in labels/logs. Precise telemetry and safe error contract are finalized in API/FN. Existing cookie transport and authorization posture retained; security design review before implementation.

Open technical ownership and next documents

Item	Owner / status
Core business rules, RP-01–05	Confirmed/approved by 006_REQ review; no open business blocker
Effective-version storage, constraints, activation, indexes/recovery	006_DB, next sequential document after BD review
Endpoint/input/response/version/error/length/date limits	006_DD-API after main DD
Exact resolution, retained history, concurrency/midnight/snapshot/budget logic	006_DD-FN
Full per-state mockups, keyboard calendar mechanics and editor detail	006_DD-SPD
Existing-screen impact addendum	Not required by current screen scope; step 10 records applicability after full designs

This is BD only. No DB/DD, application code, new dependency, test execution, commit/PR/deployment or old approved design edits. Source, SVGs and current EN/JA PDFs form this sequential review package.